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- ASD&D

* Assignment - 2

1) Devops pipeline - CI/CD pipeline setup using Github Actions / Jenkins

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A CI/CD pipeline automates the process of building, testing and deploying software of helps developers detect errors early and deliver software faster.

CI / CD pipeline.

Developer → Git Push → Build → test → Deploy

Using github actions.

Github actions can be used to automatically execute these steps. whenever code is pushed to the repository.

Steps:

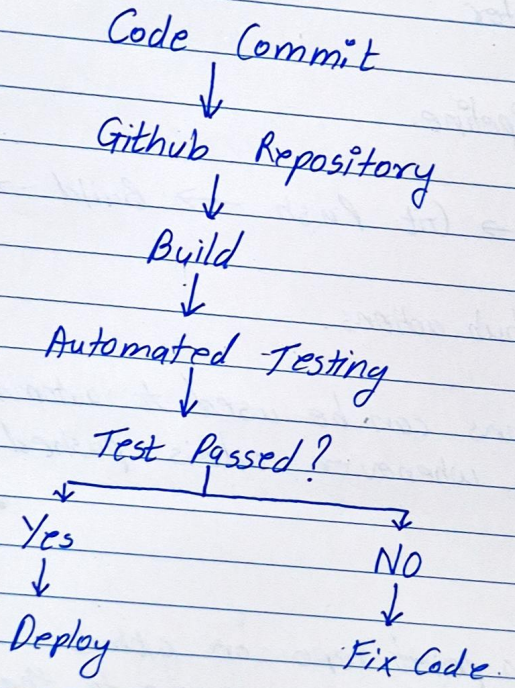
1) Create a project repo on github
2) Add the application source to the repository
3) Create a workflow file inside: github/workflow/ CI-CD
4) Configure the workflow file inside.

- Checkout the code
- Install dependencies.
- Build the applications
- Run automated tests.

Deploy application if tests puss

5) Push code to github
6) Github actions automatically starts the pipeline
7) The result of each stage can be viewed under the action tab.

Example pipeline:



Advantages:
- Automatic testing
- Faster software delivery
- Early detection of errors
- Consistent deployment.
- Reduce manual work.

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2) Simulate a full Agile process Implementation. using JIRA

→ Agile process in JIRA :

Product Backlog → Sprint Planning → Development → testing → Sprint Review → Retrospective.

Steps:
1) Create Scrum project in JIRA
2) Create project backlog: Add user stories like, login, search, order, pay, track.
3) Assign priorities
4) Create Sprint
5) Create tasks
6) Execute sprint
7) Sprint Review.
8) Sprint Retrospective.
9) Update Backlog.

Example JIRA Board.

TO-DO	In Progress	Testing	Done
Payment	Search food	login	Restaurant List
Tracking	Place Order	Cart	-

Benefits:

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Benefits:
- Early tracking of work.
- Better team collaboration
- Transparent project progress
- supports scrum ceremonies.
- ~~Keeps~~ Helps manage product Backlog.

3) Integrate Agile planning with devops automation

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Agile focuses on frequent development and delivery, while devops automates the process of building, testing and deploying software. Integrating both allows teams to ~~de~~ deliver working software continuously

Agile+ devops Flow.

Product Backlog
↓
Sprint planning
↓
Development
↓
Git Commit
↓
CI pipeline
↓
Build
↓
Test.

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↓
Deployment
↓
Monitoring
↓
Feedback
↓
Product Backlog

Integration process:
1) Agile Planning
2) Sprint planning
3) Development
4) Version control
5) CI automation
6) Continous deployment
4) monitoring
8) Feedback.
9) Next sprint

Benefits:
- Faster
- Frequent Releases
- Reduced Deployment errors.
- Quick feedback and improved software quality.